

Analytics in Business

Dr Alfonso Ferrandez

Get familiar with the materials

1 Session HSO Case Study Online Published	2 Session Session 1: Turning Numbers into Action (Week 1) Blended Published	3 Session Session 2: Exploring the Energy Industry (Week 2) Campus Published	4 Session Session 3: Design Thinking (Weeks 3 and 4) Campus Published
5 Session Session 4: Presenting with impact (Week 4) Blended ~ 1 hour Published	6 Session Session 5: Polishing the Presentations (Week 5) Blended ~ 2 hours Published	7 Session Assignment submissions Campus Published	8 Activity Reading list Campus Published

Cracking the code

Discover the key stages in the analytics process and learn how to effectively communicate your insights to both technical and non-technical audiences



“Without data, you’re
just another person
with an opinion”

W. Edwards Deming

The Analytics Journey

1

Data Discovery & Cleaning

Gather data from client, industry, market research... It will be messy, with holes and outliers.

2

Hypotheses Creation & Validation

Use intellectual intuition and your domain knowledge to propose a set of hypotheses. Prove or disprove with data.

3

Data Visualisation & Insights

Best way to understand data is to visualise it. Insights may or may not be apparent, so be creative when visualising.

4

Observations & Recommendations

Your insights will be the source of a set of recommendations that may result in shifting the metrics that need moving.

5

Actionable Insights

The most valuable output a client can get from us is a curated set of insights into their challenges and how to fix them



Cleaning up the mess

- Identify and remove **duplicates**
- Fill in **missing** data
- Dealing with **outliers**
- Correct **inconsistent** or **incorrect** data
- **Normalise** and standardise data formats

Brainstorming brilliance

Hypotheses Formation

- Use domain knowledge and **intuition**
- Observe **patterns and trends** in the data
- Consider potential factors **influencing** the outcome
- Formulate a set of **testable** hypotheses



“If you torture the
data long enough, it
will confess.”

Ronald Coase



Searching for the truth

Proving or disproving your hypotheses

- Choose appropriate statistical tests
- Determine **significance** levels
- Evaluate test results to **accept** or **reject** each hypothesis
- Consider potential **biases** and confounding factors
- **Validate your results** with additional data or experiments
- Ensure **reproducibility** of your analysis

Artful analysis

Data visualisation

- Choose the **right visualisation** type
- Focus on the **most important** data points
- Keep visuals clean and **easy to understand**
- Use colour and size to emphasize **key information**



Financial Highlights

+27%

Revenue
DKK 55.3 billion

+22%

Consumer Sales

+32%

Operating Profit
DKK 17.0 billion



Market Share
Grew in all market groups

Innovating

832 LEGO stores

(165 opened in 2021)

340 stores in China
(95 opened in 2021)

70 stores feature new immersive retail experiences

Accelerating Digital

4 digital hubs



Denmark
Billund Copenhagen
UK
China

Top-Selling Themes

LEGO® City

LEGO® Technic

LEGO® Creator Expert

LEGO® Harry Potter™

LEGO® Star Wars™

Positive Impact on

98%

increase in solar panel capacity at LEGO factories compared to 2020

3.5+ million

children reached with learning through play activities

Maximum context

The power of infographics

- **Combine** disparate data sets for context
- Use visuals to **simplify** complex information
- Tell a **cohesive story** through data
- **Engage** and educate your

Data storytelling

Tell the right story to the right audience

- **Structure** your presentation logically
- Use **visuals** to support and clarify your message
- **Simplify** and reduce jargon
- Provide context and **clear explanations**
- Focus on **actionable insights**
- Use **storytelling** techniques to engage the audience





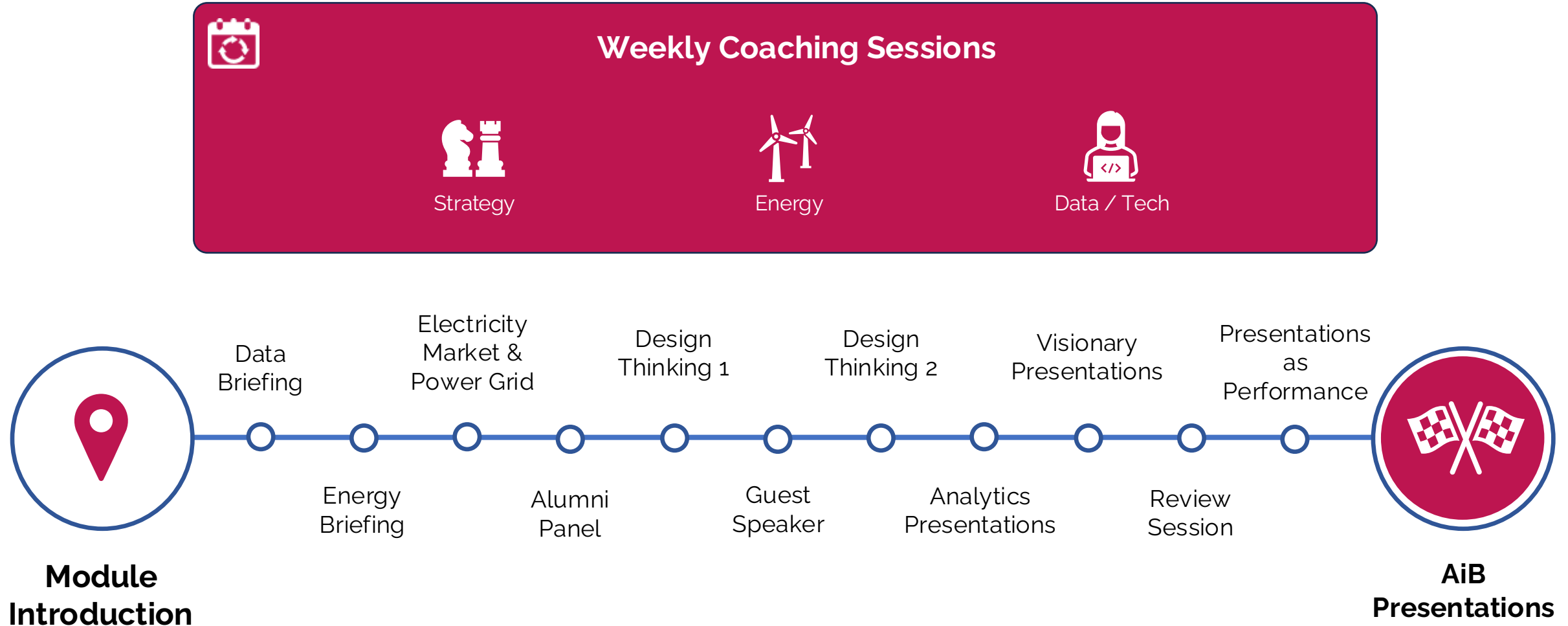
Lights, camera, action!

- Interpret and summarise **key findings**
- Identify potential **opportunities** and **risks**
- Make **data-driven** recommendations
- Communicate the **impact** and **value** of your insights

“The goal is to turn
data into information,
and information into
insight.”

Carly Fiorina, Former CEO of Hewlett-Packard

Live Sessions



Heat Smart Orkney



@TheRestlessCTO

<https://hso.indigolabs.ai>

Any questions?

